

# Daniel Zheng

Senior Software Developer

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## Summary

**Senior Software Developer** with 4+ years of experience architecting cloud-native software, public APIs, and full-stack applications for Industrial IoT platforms. Proven track record of leading technical initiatives, mentoring developers, optimizing system performance, and delivering scalable production software using **Node.js**, **React**, and **Google Cloud**.

## Education

**McMaster University**

Honours Computer Science

B.A.Sc.

2017-2022

## Experience

**Senior Software Developer**

June 2022-Present

Systems With Intelligence

- **Serve as technical lead** for the customer dashboard of a commercial Industrial IoT platform, driving architecture, code quality, and feature delivery while mentoring **2 full-time software developers** and multiple co-op students through code reviews, design discussions, and onboarding.
- **Architected and built** the company's public **IoT API** from inception, designing its **Google Cloud** architecture, **20+ secure REST API endpoints**, **API Gateway**, and **JWT** authentication, acquiring **10+ enterprise customer organizations** within the first month of release.
- **Optimized** historical Modbus telemetry visualization by implementing hierarchical time-series aggregation (hourly and daily), **reducing worst-case dashboard rendering time from ~30 seconds to under 5 seconds (~83% improvement)** while preserving drill-down access to raw measurements..
- **Modernized the customer portal** through UI/UX enhancements and delivery of high-impact features including **Access Groups** and **Label Management**, improving customer administration workflows and accelerating delivery of customer-requested enhancements.
- **Collaborated** with product, manufacturing, QA, and operations teams to rapidly deliver customer-facing features and production-critical fixes while improving release quality through automated testing, CI/CD enhancements, and engineering best practices.

**Software Developer Intern**

September 2021 – December 2021

Twins TouCans Inc.

- **Developed** internal warehouse management tools using **Django** and **Python** to support inventory tracking and fulfillment workflows.
- **Integrated** with the **Amazon Selling Partner API** to automate inventory synchronization and order processing.
- **Collaborated** with operations teams to deliver features that improved warehouse efficiency and inventory visibility.

## Projects

**AI Portfolio Website (In Progress)**

**Next.js • React • LangChain • OpenAI API • RAG • Vector Database**

- **Redesigning** and modernizing my personal portfolio website with a responsive UI and improved user experience.
- **Building** an AI-powered assistant that represents my professional persona, allowing visitors to ask natural-language questions about my experience, projects, technical skills, and career.
- **Designing** a **Retrieval-Augmented Generation (RAG)** pipeline that retrieves curated personal knowledge from external documents to generate accurate, context-aware responses while reducing hallucinations.

- Exploring vector embeddings, semantic search, prompt engineering, and conversational memory to create a personalized AI experience.

## Technical Skills

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**Languages :** JavaScript • Python • Java • C/C++

**Frontend:** React • Next.js • React Native

**Backend:** Node.js • Express • REST APIs

**Cloud:** Google Cloud • Docker • Git • Bitbucket Pipelines

**Database:** MySQL • Datastore